

Trainings ▾

Fuel Recommendations

CAUTION

Because of precise tolerances of diesel injection systems, it is extremely important that the fuel be kept clean and free of dirt or water. Dirt or water in the system can cause severe damage to both the injection pump and the injection nozzles.

CAUTION

Do not use diesel fuel blended with lubricating oil in engines equipped with a catalytic converter (including 1994 model year and beyond). Damage to legally required emission control can result.

Use ASTM Number 2 D fuel with a minimum cetane number of 40. Number 2 diesel fuel gives the best economy and performance under most operating conditions. Fuels with cetane numbers higher than 40 can be required in higher altitudes or extremely low ambient temperatures to prevent misfires and excessive smoke.

At operating temperatures below 0°C [32°F], use a blend of Number 1 D and Number 2 D fuels, also known as “winterized” Number 2 D.

Note : Number 1 D fuel can be used; however, fuel economy will suffer.

Use low-sulfur content fuel having a cloud point that is at least 10 degrees below the lowest expected fuel temperature. Cloud point is the temperature at which wax crystals begin to form in diesel fuel.

The viscosity of the fuel **must** be kept above 1.3 centistokes to provide adequate fuel system lubrication at 40°C [104°F].

For a more detailed description of fuel properties, refer to Fuel for Cummins Engines, Bulletin 3379001 (</qs3/pubsys2/xml/en/bulletin/3379001.html>).

The following chart lists acceptable alternate fuels for midrange engines.

CAUTION

Do not use diesel fuel blended with lubricating oil in engines equipped with a catalytic converter (including model year 1994 and beyond). Damage to legally required emission control can result.

Note : Any adjustment to compensate for reduced performance with a fuel system using alternate fuel is **not** warrantable.

Note : Wear on any midrange fuel pump component attributed to the lack of lubrication in the fuel is **not** a warrantable repair.

Acceptable Alternate Fuels - Component Wear/Durability			
Fuel Type	Bosch® A Pump	Bosch® P Pump	Bosch® VE Pump
Number 1-D diesel	OK	OK	OK
Number 2-D diesel	OK	OK	OK
Number 1-K kerosene	OK	OK	*
Number 2-K kerosene	OK	OK	*
Jet-A	OK	OK	*
Jet A-1	OK	OK	*
JP-5	OK	OK	*
JP-8	OK	OK	*
Jet-B	Not OK	Not OK	Not OK
JP-4	Not OK	Not OK	Not OK
Cite	Not OK	Not OK	Not OK
* OK only if 5 percent new lubricating oil is blended with these fuels to increase the lubricity to acceptable level.			
Acceptable Alternate Fuels - Component Wear/Durability			
Fuel Type	Denso EP-9	Stanadyne DB-4	Lucas CAV (DPA and DPS)
Number 1-D diesel	OK	OK	OK
Number 2-D diesel	OK	OK	OK
Number 1-K kerosene	OK	*	*
Number 2-K kerosene	OK	*	*
Jet-A	OK	*	*
Jet A-1	OK	*	*
JP-5	OK	*	*
JP-8	OK	*	*
Jet-B	Not OK	Not OK	Not OK
JP-4	Not OK	Not OK	Not OK
Cite	Not OK	Not OK	Not OK
* OK only if 5-percent new lubricating oil is blended with these fuels to increase the lubricity to acceptable level.			

Fuel Filters

A = Standard filter used as secondary filter in dual-filter applications

B = Fuel-water separator primary filter for dual-filter applications

C = Fuel-water separator used in single-filter applications.

